

Luo Yan

✉ luoyan99@sjtu.edu.cn

☎ (+86)188-174-64668

Supervisor: Ding Han (Academician of the Chinese Academy of Sciences),

 **Xiong Zhenhua** (Professor at Shanghai Jiao Tong University).



EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China

Sep.2021-June.2026

Major in Mechanical Engineering, going from a Bachelor's to a Ph.D.

Relevant Coursework: Computational vision (A), Digital signal processing(A), Robot performance simulation and control principle(A-), Basis of software technique(A), Modern control theory(A-), Intelligent control technique(A-). **(GPA:3.51/4.0)**

Yanshan University (YSU), Hebei, China

Sep.2017-June.2021

Major in Automation, Bachelor. (Ranked 1st out of 373 students).

Relevant Coursework: Advanced Mathematics(B+), Linear Algebra(A), Feedback Control Theory(A), Microcomputer Principle(A+), Simulation Electronic Technology(A), Physical Fitness Test(A), Embedded single chip microcomputer principle(A+), Numerical Analysis(A).



EXPERIENCE AND SKILLS

- Responsible for intelligent manipulation and grasping of semi-humanoid robotic embodiment, including **language-vision-motion modeling**. Using **teleoperation** to acquire motion data combined with **reinforcement learning** for training.
- Responsible for Jiangsu Provincial Science and Technology Department's Frontier Leading Technology Basic Research Program and Jiangsu Province Industrial And Information Industry Transformation And Upgrading Project : **Humanoid robot** dynamic balance control algorithm research and development.
- Participated in the project of Science and Technology Development Program of Jilin Province: Research and development of robot operation system based on machine vision technology, responsible for **3D vision** disordered grasping and **vision servo**.
- Participated in China Internet+ Innovation and Entrepreneurship Competition, responsible for **quadruped robot** control algorithm development as well as China University Student Mathematical Modeling Competition, responsible for **simulation and algorithm optimization**.
- Proficient in Python, C++, C, Matlab, Blender, SolidWorks, etc. Familiar with Ubuntu, ROS, Gazebo, MuJoCo, Isaac Gym/Sim/Lab, etc., with strong self-driven learning ability.



ACADEMIC ACHIEVEMENTS

- **Luo Y**, Xiong Z, Ding H, et al. CoPickVLM: A Vision-Language Model Guided Dual-Arm Collaborative System for OcclusionAware Tomato Harvesting. (Under Review) Mar.2026
- **Luo Y**,Xiong Z, et al. MotionVL:Visual-Language Supervision for Guiding Reinforcement Learning in Humanoid Motion[J].IEEE Robotics and Automation Letters. Feb.2026
- **Luo Y**, Liu C, Xiong Z, et al. DCM-based dynamic stable walking under terrain-induced

- time-varying disturbances for humanoid robots[J]. Science China Tech Sciences. *Jan. 2025*
- Pang F, Chen Y, **Luo Y**, et al. A Fast Obstacle Detection Algorithm Based on 3D LiDAR and Multiple Depth Cameras for Unmanned Ground Vehicles[J]. Drones. *Nov.2024*
 - **Luo Y**. Technical Requirements for Humanoid Robotics (**T/CIET 648–2024**). China International Economic and Technical Cooperation Promotion Association Standard. *Sep. 2024*
 - **Luo Y**, Chen G, Liu C, et al. Image Foreground Segmentation Based on Small Data Set for Visual Servo Applications[C]//2023 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM). IEEE, 2023: 715-720. *Jun.2023*
 - Chen G, **Luo Y**, Huang B, et al. BP neural network-fuzzy control-based uncalibrated visual servoing technology for robots[J]. Science, Technology and Engineering, 2023. *Mar.2023*
 - Zhang M, **Luo Y**, Chen G, et al. Design and research of intelligent maintenance robot for steam generator in nuclear power plant[J]. Science, Technology and Engineering, 2023, 23(22): 9559-9566. *Feb,2023*
 - [Invention Authorization] A Method for Remote Operation Control of a Semi-Humanoid Robot Based on VR Devices (ranking second) *Nov.2025*
 - [Invention Authorization] A visual servo method based on the mixing of multiple image feature information (ranking second) *Oct.2022*
 - [Invention Announcement] A cross-modal alignment visual language navigation method in dynamic occlusion environment (ranking third) *Dec.2025*



HONORS AND AWARDS

- Outstanding Graduate, Shanghai Jiao Tong University(SJTU) *Dec.2025*
- Selected for the Jiangsu Province Young Talent Support Program *Jul.2025*
- Senior Consultant Expert of Nanjing Production and Economy Consultant Pool *Oct.2024*
- Chief liaison officer of world famous universities of Jiangsu Innovation and Entrepreneurship Alliance *Apr.2024*
- Academic first-class scholarship of SJTU *Dec.2025&2024&2023&2022*
- Yanshan University Person of the Year (Only 5 person in school) *Jan.2021*
- National scholarship (award percentage was 0.2%) *Dec.2020*
- Hebei provincial Merit Student (award percentage was 0.14%) *May.2020*
- National Bronze Award in the 5th “Internet+” Innovation and Entrepreneurship Competition (ranking third) *Dec.2019*
- First Prize of Hebei Division of National College Students Mathematical Modeling Competition (ranking first) *Oct.2019*



ACTIVITIES

- Invited by the **Chinese Association of Automation**, I delivered a technical presentation at the World Intelligent Manufacturing Conference. I engaged in in-depth discussions on humanoid robot industrialization with **Xingxing Wang, CEO of Unitree Robotics**, and **chairpersons of publicly listed companies**, and presented research outcomes to the **Governor of Jiangsu Province**. I also presented my work at academic conferences AIM in Seattle, USA. Will attend the IROS 2026 International Conference in Pittsburgh, USA.